(Hopefully) Understanding Efficient Influence Functions

Semiparametric Efficiency Reading Group
Session 1

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Outline

1. Problem Setting
2. Influence Functions & Efficient Influence Functions
3. Building Efficient Estimators
   - One-Step Estimator
   - Estimating Equation
   - Targeted Learning
4. Asymptotical Analysis of the One-Step Estimator
5. Local Minimax Optimality in Nonparametric Efficiency Bounds

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Problem Setting: Notation and Target Parameter

We observe independent samples $Z_1, \ldots, Z_n \in \mathcal{Z}$ drawn from an unknown distribution $\mathbb{P}$, assumed to lie in a statistical model $\mathcal{P}$.

Empirical and population averages:

- $\mathbb{P}_n(f) = \frac{1}{n} \sum_{i=1}^{n} f(Z_i)$
- $\mathbb{P}(f) = \int f(z)\, d\mathbb{P}(z)$
- $\|f\|^2_{L_2(\mathbb{P})} = \int f(z)^2\, d\mathbb{P}(z)$

Goal

Estimate a *target parameter* (functional)

$$\psi : \mathcal{P} \longrightarrow \mathcal{D},$$

$$P \longmapsto \psi(P),$$

where $\mathcal{D}$ is typically $\mathbb{R}$.

Examples of Target Parameters

Example

Regression Function: Let $Z = (X, Y)$. The regression of Y on X is

$$\psi_x(P) = \mathbb{E}_P(Y \mid X = x),$$

where $\mathcal{D}$ is the set of functions from $\mathbb{R}^d$ to $\mathbb{R}$.

Example

Average Treatment Effect: Let $Z = (X, T, Y)$, where X are observed confounders, T is treatment, and Y is the outcome. Under standard causal assumptions, the mean outcome if treated at level $T = t$ is

$$\psi_t(P) = \mathbb{E}_P\{Y^{(t)}\} = \mathbb{E}\{\mu_t(X)\},$$

with $\mu_t(X) = \mathbb{E}(Y \mid X, T = t)$.

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Von Mises Expansion

Definition

Gâteaux derivative Let $\psi : \mathcal{P} \to \mathbb{R}$ be a smooth functional. The Gâteaux derivative of ψ at P in direction H is

$$\psi'(P)[H] = \frac{d}{dt}\,\psi(P + tH)\Big|_{t=0}.$$

Von Mises expansion. For distributions P and Q sufficiently close,

$$\psi(Q) = \psi(P) + \psi'(P)[\,Q - P\,] + R_2(P, Q),$$

where $R_2(P, Q)$ is a second-order remainder, analogous to the remainder in a Taylor expansion.

Interpretation.

- $\psi'(P)[Q - P]$ captures the *first-order* effect of perturbing P toward Q.
- If this first-order term can be represented as an $L_2(P)$ inner product, the representing function will later be the *influence function*.
- We will revisit this representation after introducing parametric submodels and the tangent space.

Parametric Submodels and Scores

Parametric submodel

A parametric submodel is a smooth family

$$\{P_\epsilon : \epsilon \in \mathbb{R}\} \subseteq \mathcal{P}$$

satisfying $P_{\epsilon=0} = P$. It provides a controlled way to perturb the data law in order to study how the parameter $\psi(P)$ changes.

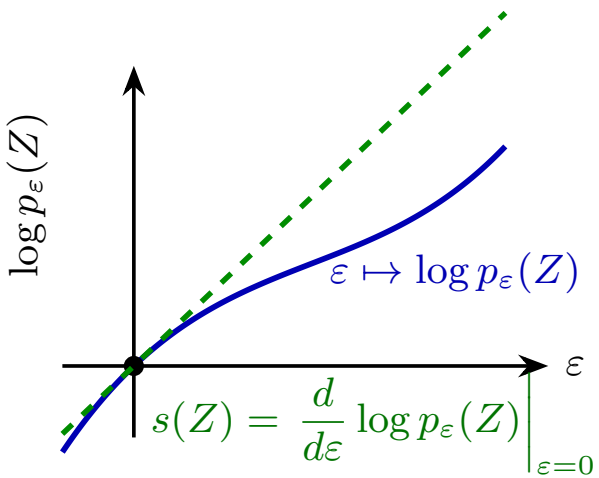
Example (nonparametric model).
A common perturbation is

$$p_\epsilon(z) = p(z)\{1+\epsilon h(z)\}, \quad \|h\|_\infty \le M,\ \epsilon < 1/M.$$

Score. The score is the infinitesimal direction of change:

$$s(z) = \frac{\partial}{\partial \epsilon} \log p_\epsilon(z)\Big|_{\epsilon=0}.$$

It belongs to $L_2(P)$, has mean zero, and represents a valid direction in which the distribution can change.



Pathwise Differentiability

Parameter path. A smooth submodel $\{P_\varepsilon\}$ induces a path

$$\varepsilon \mapsto \psi(P_\varepsilon),$$

describing how the parameter changes when the data distribution is perturbed.

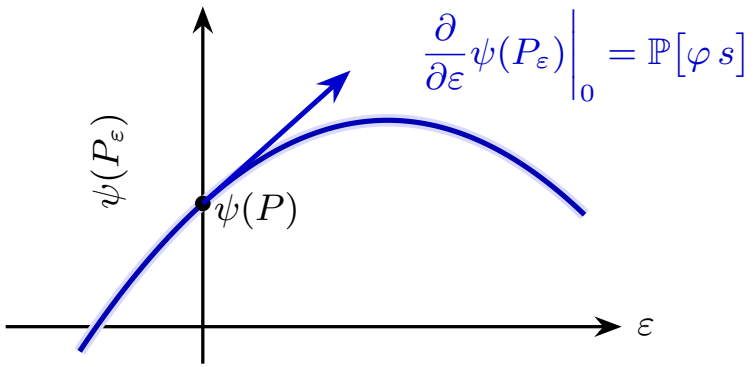
Pathwise derivative

The parameter ψ is *pathwise differentiable* at P if, for every smooth submodel $\{P_\varepsilon\}$ through P, the limit

$$\frac{d}{d\varepsilon}\,\psi(P_\varepsilon)\Big|_{\varepsilon=0}$$

exists.

Interpretation. Pathwise differentiability ensures that $\psi(P)$ has a well-defined first-order response to smooth perturbations of the distribution.



The Tangent Space

Motivation. Smooth submodels describe local perturbations of P. Their scores provide the *directions* along which the pathwise derivative $\frac{d}{d\varepsilon}\psi(P_\varepsilon)\big|_0$ is taken.

Scores of parametric submodels

For any smooth submodel $\{P_\epsilon\}$ passing through P, the score at $\epsilon = 0$ is

$$s(z) = \frac{\partial}{\partial \epsilon} \log p_\epsilon(z)\Big|_0 .$$

Each score is a mean-zero element of $L_2(P)$ and represents a valid infinitesimal "direction" in which P can change.

Tangent space

The *tangent space* at P is the $L_2(P)$-closure of all such score directions:

$$\mathcal{T}(P) = \overline{\{\, s(z) : s \text{ is the score of a smooth submodel through } P \,\}}^{L_2(P)}.$$

In a fully nonparametric model, $\mathcal{T}(P)$ is the space of all mean-zero, finite-variance functions in $L_2(P)$.

Pathwise Derivative as a Linear Functional

Directional derivative. For any smooth submodel $\{P_\varepsilon\}$ with score $s \in \mathcal{T}(P)$,

$$\psi'(P)[s] = \left.\frac{d}{d\varepsilon}\psi(P_\varepsilon)\right|_{\varepsilon=0}.$$

Why this map is linear. If s_1 and s_2 are scores of two submodels, then for any $a, b \in \mathbb{R}$ the combined perturbation

$$p_\varepsilon(z) = p(z)\{1 + \varepsilon(as_1(z) + bs_2(z))\}$$

defines a valid submodel with score $as_1 + bs_2$.
Applying the chain rule to the bivariate path $\psi(P_{\varepsilon a,\, \varepsilon b})$,

$$\psi'(P)[as_1 + bs_2] = a\,\psi'(P)[s_1] + b\,\psi'(P)[s_2].$$

Thus the map

$$L : \mathcal{T}(P) \to \mathbb{R}, \qquad L(s) = \psi'(P)[s],$$

is a **linear functional** on the tangent space.

Efficient Influence Function

Linear functional on the tangent space. The pathwise derivative defines

$$L : \mathcal{T}(P) \to \mathbb{R}, \qquad L(s) = \psi'(P)[s].$$

Efficient influence function (Riesz representer)

If L is bounded on the Hilbert space $\mathcal{T}(P)$, the Riesz representation theorem guarantees a unique

$$\varphi \in \mathcal{T}(P) \quad \text{such that} \quad L(s) = \int \varphi(z; P)\, s(z)\, dP(z) \quad \forall s \in \mathcal{T}(P).$$

This φ is the **efficient influence function** (canonical gradient) of ψ.

Remarks.

- Boundedness of L is the key condition for existence of φ; when it fails, the parameter is *not* pathwise differentiable.
- The EIF is the unique element of $\mathcal{T}(P)$ that encodes the first-order behavior of ψ under any smooth perturbation of P.

Efficient Influence Function as Projection

Influence functions. A function $h(Z) \in L_2(P)$ is a (regular) influence function if it reproduces the pathwise derivative along every score direction:

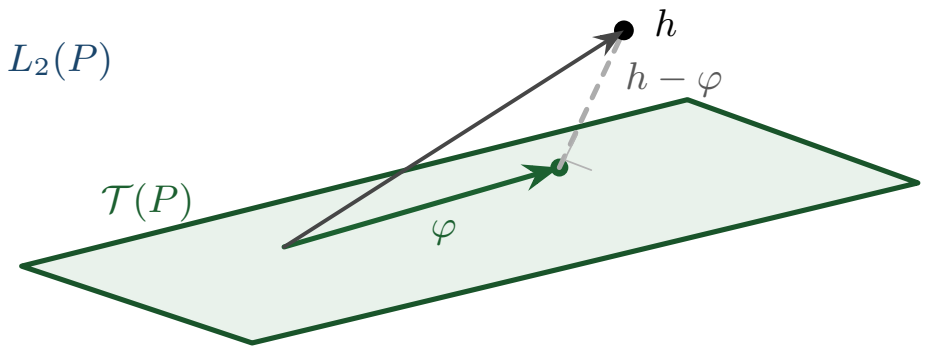
$$\frac{\partial}{\partial \varepsilon} \psi(P_\varepsilon)\Big|_0 = \mathbb{E}[h\, s] \qquad \forall\, s \in \mathcal{T}(P).$$

Efficient Influence Function (EIF)

Among all such h, the EIF φ is the *unique element of the tangent space* $\mathcal{T}(P)$ satisfying

$$\mathbb{E}[\varphi\, s] = \mathbb{E}[h\, s] \qquad \forall\, s \in \mathcal{T}(P),$$

and having *minimal* $L_2(P)$ norm.



Efficiency. The EIF φ is the projection of any valid influence function h onto $\mathcal{T}(P)$. Efficient estimators are those whose asymptotic linear expansion uses the EIF.

Pathwise Differentiability and the Efficient Influence Function

Von Mises expansion along a submodel. For a smooth submodel P_ε with score

$$s(z) = \left.\frac{\partial}{\partial \varepsilon} \log p_\varepsilon(z)\right|_0,$$

we have

$$\psi(P_\varepsilon) = \psi(P) + \varepsilon \int \varphi(z; P)\, s(z)\, dP(z) + R_2(P_\varepsilon, P),$$

with

$$\left.\frac{\partial}{\partial \varepsilon} R_2(P_\varepsilon, P)\right|_0 = 0.$$

Implication. Because the remainder's first derivative vanishes,

$$\left.\frac{d}{d\varepsilon}\psi(P_\varepsilon)\right|_0 = \int \varphi(z; P)\, s(z)\, dP(z).$$

Interpretation.

- The derivative defines a linear functional acting on score directions s.
- Pathwise differentiability means this linear functional has an $L_2(P)$ representer φ.
- φ captures the *first-order sensitivity* of ψ to smooth perturbations of P.

Existence of the Influence Function

Pathwise differentiability and existence.

- The pathwise derivative $\psi'(P)[H]$ acts on the tangent space $\mathcal{T}(P) \subseteq L_2(P)$.
- If $\psi'(P)[H]$ is a **bounded linear functional**, the Riesz representation theorem guarantees a unique

$$\varphi \in \mathcal{T}(P) \quad \text{s.t.} \quad \psi'(P)[H] = \int \varphi(z; P)\, h(z)\, dP(z),$$

  where $h = dH/dP$.
- Therefore, existence of φ is **equivalent** to **pathwise differentiability** of ψ (Bickel et al., 1993; Kennedy, 2024).

When the influence function does not exist:

- The functional is not Gateaux/Fréchet differentiable in directions of the tangent space.
- Examples:
  - Density at a point, quantiles (Tsybakov, 2009)
  - Non-smooth causal functionals: max/min-type parameters
  - Optimal treatment rule value without margin conditions (Hirano et al., 2012)

Further Remarks on Influence Functions

Multiple names for φ:

- Influence function / influence curve (Bickel et al., 1993; Kennedy, 2024)
- Pathwise derivative / pathwise gradient (Newey, 1994)
- Neyman orthogonal score (Chernozhukov et al., 2018)

Important distinctions:

- The *influence curve* is a mathematical property of the parameter ψ: it is the unique function that represents the pathwise derivative of ψ in the model, independent of any estimator.
- The *influence function of an estimator* is the function that appears in its asymptotic linear expansion. Different estimators generally have different influence functions, and not all of them equal the efficient influence curve.
- Efficient estimators (one-step, TMLE, DML) are precisely those whose asymptotic influence function equals the model's efficient influence curve.

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Deriving the Influence Function: Pathwise Derivative

Using the pathwise derivative formula, we can compute influence functions.

Example

Regression: The influence function for $\mathbb{E}(Y \mid X = x)$ using the pathwise derivative is

$$\varphi(z; P) = \frac{1(X = x)}{P(X = x)} \big[Y - \mathbb{E}_P(Y \mid X = x) \big].$$

Pathwise Derivative for $\mathbb{E}(Y \mid X = x)$

We consider the functional

$$\psi(P) = \mathbb{E}_P[Y \mid X = x] = \frac{\mathbb{E}_P\left[Y\, 1(X = x)\right]}{P(X = x)}.$$

Let $\{P_\varepsilon\}$ be any smooth parametric submodel with score

$$s(z) = \left.\frac{\partial}{\partial \varepsilon} \log p_\varepsilon(z)\right|_0.$$

Step 1: Differentiate numerator and denominator. Let

$$N_\varepsilon = \mathbb{E}_\varepsilon[Y\, 1(X = x)], \qquad D_\varepsilon = P_\varepsilon(X = x).$$

Then

$$\psi(P_\varepsilon) = \frac{N_\varepsilon}{D_\varepsilon}.$$

By standard pathwise calculus,

$$\left.\frac{\partial}{\partial \varepsilon} N_\varepsilon\right|_0 = \mathbb{E}_P[Y\, 1(X = x)\, s(Z)],$$

$$\left.\frac{\partial}{\partial \varepsilon} D_\varepsilon\right|_0 = \mathbb{E}_P[1(X = x)\, s(Z)].$$

Step 2: Quotient rule.

$$\left.\frac{d}{d\varepsilon}\psi(P_\varepsilon)\right|_0 = \frac{\mathbb{E}[Y\, 1(X = x)s(Z)] - \psi(P)\ \mathbb{E}[1(X = x)s(Z)]}{P(X = x)}.$$

Influence Function for $\mathbb{E}(Y \mid X = x)$

From the previous slide,

$$\psi'(P)[s] = \frac{\mathbb{E}[Y\,1(X = x)s(Z)] - \psi(P)\;\mathbb{E}[1(X = x)s(Z)]}{P(X = x)}.$$

Rewrite the numerator as a conditional expectation:

$$\mathbb{E}[Y\,1(X = x)s(Z)] = P(X = x)\;\mathbb{E}[Ys(Z) \mid X = x].$$

Similarly,

$$\mathbb{E}[1(X = x)s(Z)] = P(X = x)\;\mathbb{E}[s(Z) \mid X = x].$$

Thus the pathwise derivative becomes

$$\psi'(P)[s] = \mathbb{E}\left[\frac{1(X = x)}{P(X = x)}\left(Y - \mathbb{E}[Y \mid X = x]\right)s(Z)\right].$$

Identification. This shows the linear functional $s \mapsto \psi'(P)[s]$ is represented by

$$\varphi(z; P) = \frac{1(X = x)}{P(X = x)}\Big(Y - \mathbb{E}_P[Y \mid X = x]\Big),$$

so that

$$\psi'(P)[s] = \mathbb{E}\left[\varphi(Z; P)\,s(Z)\right].$$

Conclusion. $\varphi(z; P)$ is the influence function for $\mathbb{E}(Y \mid X = x)$.

Practical Tricks for Computing Influence Functions

Define an operator $\mathbb{IF} : \Psi \to L_2(P)$ that maps functionals $\psi : \mathcal{P} \to \mathbb{R}$ to their influence functions $\varphi(z) \in L_2(P)$.

Tricks:

1. Pretend the data are discrete.
2. Treat influence functions as derivatives (apply differentiation rules).
3. Use building blocks:

$$\mathbb{IF}(\mathbb{E}(Y \mid X = x)) = \frac{\mathbf{1}(X = x)}{P(X = x)}\{Y - \mathbb{E}(Y \mid X = x)\}$$

$$\mathbb{IF}(\mathbb{P}(X = x)) = \mathbf{1}(X = x) - \mathbb{P}(X = x)$$

1. **Product Rule:**
$$\mathbb{IF}(\psi_1\psi_2) = \mathbb{IF}(\psi_1)\psi_2 + \psi_1\mathbb{IF}(\psi_2)$$
2. **Chain Rule:**
$$\mathbb{IF}(f(\psi)) = f'(\psi)\mathbb{IF}(\psi)$$

Example

Average Treatment Effect (ATE)

$$\psi_t = \mathbb{E}[\mu_t(X)], \quad \mu_t(X) = \mathbb{E}(Y \mid X, T = t), \quad \pi_t(X) = \mathbb{P}(T = t \mid X).$$

$$\mathbb{IF}(\psi_t) = \frac{\mathbf{1}(T = t)}{\pi_t(X)}(Y - \mu_t(X)) + \mu_t(X) - \psi_t.$$

Building Estimators

Objective: Estimate $\psi(\mathbb{P})$ using an estimated distribution $\widehat{\mathbb{P}}$.
Von Mises Expansion:

$$\psi(\widehat{\mathbb{P}}) - \psi(\mathbb{P}) = -\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z) + R_2(\widehat{\mathbb{P}},\mathbb{P})$$

The plug-in estimator $\hat{\psi}_{pi} = \psi(\widehat{\mathbb{P}}))$ has first-order bias: $\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z)$

Example

Average Treatment Effect (ATE) Plug-in estimator:

$$\widehat{\psi}_{pi} = \mathbb{P}_n\{\widehat{\mu}_t(X)\}, \quad \mu_t(x) = \mathbb{E}(Y \mid X = x, T = t).$$

Bias:

$$\mathbb{E}(\psi - \widehat{\psi}_{pi}) = \int \mathbb{E}\{\mu_t(x) - \widehat{\mu}_t(x)\}d\mathbb{P}(x).$$

Debiasing Strategies:

1. **One-step correction:** Estimate $\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z)$ directly.
2. **Estimating equation:** Build an estimator to cancel $\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z)$.
3. **Targeted learning:** Adjust nuisance parameter to cancel $\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z)$.

One-Step Estimator

The first estimator $\hat{\psi}^{(os)}$ is the **one-step estimator** or bias correction estimator (Kennedy, 2024), which estimates the bias term $-\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z)$ and subtracts it off. A natural estimator is the corresponding sample average:

$$-\mathbb{P}_n\left\{\varphi(Z,\widehat{\mathbb{P}})\right\},$$

leading to the one-step estimator:

$$\hat{\psi}^{(os)} = \psi\left(\widehat{\mathbb{P}}\right) + \mathbb{P}_n\left\{\varphi\left(Z;\widehat{\mathbb{P}}\right)\right\}.$$

Example

Average Treatment Effect (ATE) Noting $\psi_1 = \mathbb{E}\{\mu_t(X)\}$, we have:

$$\varphi(Z;\mathbb{P}) = \frac{\mathbf{1}(T=t)}{\mathbb{P}(T=t \mid X)}(Y-\mu_t(X)) + \mu_t(X) - \psi.$$

Thus, the one-step estimator is:

$$\hat{\psi}^{(os)} = \hat{\psi}_{pi} + \mathbb{P}_n\{\varphi(Z,\mathbb{P}_n)\} = \sum_{i\in\mathcal{I}}\left(\hat{\mu}_t(X_i) + \mathbf{1}(T_i=t)\frac{Y_i-\hat{\mu}_t(X_i)}{\hat{\pi}_t(X_i)}\right).$$

Estimating Equation Technique

The **estimating equation estimator** identifies a value for $\hat{\psi}$ and nuisance parameters $\hat{\eta}$ associated to $\widehat{\mathbb{P}}$ such that:

$$\mathbb{P}_n \varphi\left(Z; \widehat{\mathbb{P}}\right) = 0.$$

Procedure:

1. Rewrite the influence function as $\varphi\left(Z_i; \psi, \eta\right)$.
2. Estimate $\hat{\eta}$ using a flexible method.
3. Determine $\hat{\psi}^{(ee)}$ such that: $\mathbb{P}_n \varphi\left(Z; \hat{\psi}^{(ee)}, \hat{\eta}\right) = 0$.

Example

Average Treatment Effect (ATE)

- Target parameter: $\psi = \mathbb{E}\{\mu_t(X)\}$.
- Influence function:

$$\varphi(Z; \psi, \mu_t, \pi) = \mathbf{1}(T = t)\frac{(Y - \mu_t(X))}{\pi_t(X)} + \mu_t(X) - \psi.$$

- Solution: $\hat{\psi} = \mathbb{P}_n \left\{\mathbf{1}(T = t)\frac{(Y - \hat{\mu}_t(X))}{\hat{\pi}_t(X)} + \hat{\mu}_t(X)\right\}$.

Most of the time, the estimating equation approach and the bias correction approach give the same estimator.

Targeted Learning

Targeted learning (TL), or Targeted Maximum Likelihood Estimation (TMLE), refines the plug-in estimator $\hat{\psi}_{pi}$ by adjusting the distribution $\widehat{\mathbb{P}}$ to a **fluctuated** version $\widehat{\mathbb{P}}^*$, correcting bias on the distributional scale.
TL involves:

1. Estimating a nuisance parameter $\widehat{\eta}$.
2. Constructing a fluctuated nuisance estimate $\widehat{\eta}^*$ such that:

$$\mathbb{P}_n \varphi \left(Z; \hat{\psi}_{pi}, \hat{\eta}^* \right) = 0.$$

3. Defining the TMLE estimator as:

$$\hat{\psi}^{tl} = \psi(\widehat{\mathbb{P}}^*) = \mathbb{P}_n\{\widehat{\eta}^*\}.$$

This process ensures that the estimator is asymptotically efficient, double-robust, and bounded, improving performance in finite-sample settings.

Application to Average Treatment Effect (ATE)

For ATE, the parameter of interest is $\psi = \mathbb{E}\{\mu_t(X)\}$, with the efficient influence function:

$$\varphi(Z; \psi, \mu_t, \pi) = \mathbf{1}(T = t)\frac{(Y - \mu_t(X))}{\pi_t(X)} + \mu_t(X) - \psi.$$

The TL procedure involves:

1. Estimating μ_t and π_t using initial models, yielding $\hat{\mu}_t$ and $\hat{\pi}_t$.
2. Updating these estimates through a fluctuation model to construct $\hat{\mu}_t^*$.

The fluctuation ensures that the updated estimates satisfy:

$$\mathbb{P}_n \left\{ \mathbf{1}(T = t)\frac{(Y - \hat{\mu}_t^*(X))}{\hat{\pi}_t(X)} + \hat{\mu}_t^*(X) - \hat{\psi} \right\} = 0.$$

Obtaining the fluctuated nuisance parameter $\hat{\mu}_t^*$

The fluctuated nuisance parameter $\hat{\mu}_t^*$ is obtained using a fluctuation model:

$$\mathcal{M}(\bar{\mu}_t, \bar{\pi}_t) = \left\{ \bar{\mu}_t^\epsilon(x) = \text{expit}\left(\text{logit}(\bar{\mu}_t(x)) + \epsilon \frac{\mathbf{1}(T = t)}{\bar{\pi}_t(x)} \right), \text{ for } \epsilon \in \mathbb{R} \right\}.$$

Key steps:

1. ϵ_n is chosen to minimize the empirical risk:

$$\epsilon_n = \underset{\epsilon \in \mathbb{R}}{\arg\min}\, \mathbb{P}_n \left(-\mathcal{L}(\hat{\mu}_t^\epsilon) \right),$$

   where:

$$-\mathcal{L}(\mu_t) = Y \log(\mu_t(X)) + (1 - Y) \log(1 - \mu_t(X)).$$

2. The fluctuated estimate is defined as $\hat{\mu}_t^* \doteq \hat{\mu}_t^{\epsilon_n}$.

This fluctuation improves the fit while maintaining consistency with the efficient influence function equation.

Proving the Efficient Influence Curve Equation

To justify $\hat{\mu}_t^*$, we consider two key results:

1. Further fluctuations do not change $\hat{\mu}_t^*$, as the derivative of the empirical risk at ϵ_n is zero:
$$\frac{d}{d\epsilon}\mathbb{P}_n(\mathcal{L}(\hat{\mu}_t^{\epsilon}))\big|_{\epsilon=\epsilon_n} = 0.$$
2. The derivative of the loss with respect to ϵ satisfies:
$$\frac{d}{d\epsilon}\,\mathbb{P}_n\left(\mathcal{L}(\mu_t^{*,\epsilon})\right)\big|_{\epsilon=0} = -\frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{1}(T_i = t)}{\hat{\pi}_t(X_i)}\left(Y_i - \hat{\mu}_t^*(X_i)\right).$$

Equivalently, this implies:

$$\mathbb{P}_n\left\{\mathbf{1}(T = t)\frac{(Y - \hat{\mu}_t^*(X))}{\hat{\pi}_t(X)}\right\} = 0.$$

By definition of the TL estimator:

$$\mathbb{P}_n\left\{\hat{\mu}_t^*(X) - \hat{\psi}\right\} = 0.$$

These results confirm that ψ^{tl} and $\hat{\mu}_t^*$ satisfy the efficient influence curve equation.

Advantages of TL

TMLE offers several advantages:

- **Finite-Sample Robustness**: Ensures $\psi(\mathbb{P})$ and $\psi(\widehat{\mathbb{P}}^*)$ remain bounded (e.g., $[-1, 1]$).
- **Efficiency**: Guarantees compatibility with the efficient influence curve equation, improving asymptotic efficiency.
- **No Iteration Required**: Once the fluctuation is performed, further updates do not alter the result.

These properties make TL a powerful tool for causal inference and semi-parametric estimation.

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Asymptotical Analysis of the One-Step Estimator

Recall the **Von Mises Expansion** and the **one-step** definition:

$$\psi(\widehat{\mathbb{P}}) - \psi(\mathbb{P}) = -\int \varphi(z;\widehat{\mathbb{P}})d\mathbb{P}(z) + R_2(\widehat{\mathbb{P}},\mathbb{P})$$

$$\hat{\psi}^{(os)} = \psi\left(\widehat{\mathbb{P}}\right) + \mathbb{P}_n\left\{\varphi\left(Z;\widehat{\mathbb{P}}\right)\right\}.$$

Key Idea: The one-step estimator can be decomposed as:

$$\widehat{\psi}^{os} - \psi = \underbrace{(\mathbb{P}_n - \mathbb{P})\{\varphi(Z;\mathbb{P})\}}_{S^*} + \underbrace{(\mathbb{P}_n - \mathbb{P})\left\{\varphi(Z;\widehat{\mathbb{P}}) - \varphi\left(Z;\mathbb{P}\right)\right\}}_{\text{Empirical Process}} + \underbrace{R_2\left(\widehat{\mathbb{P}},\mathbb{P}\right)}_{\text{Second-Order Remainder}}.$$

Components:

- **Sample Average (S^*)**: $(\mathbb{P}_n - \mathbb{P})\{\varphi(Z;\mathbb{P})\}$
  - Behaves as a normally distributed random variable with variance $\text{var}(\varphi)/n$, up to $o_{\mathbb{P}}(1/\sqrt{n})$.
- **Empirical Process**: $(\mathbb{P}_n - \mathbb{P})\left\{\varphi(Z;\widehat{\mathbb{P}}) - \varphi\left(Z;\mathbb{P}\right)\right\}$
- **Second-Order Remainder**: $R_2\left(\widehat{\mathbb{P}},\mathbb{P}\right) = \left[\psi(\widehat{\mathbb{P}}) - \psi\left(\mathbb{P}\right)\right] - \mathbb{P}\left\{\varphi\left(Z;\widehat{\mathbb{P}}\right)\right\}$.

Empirical Process and L_2 Consistency

Key Term: The empirical process term:

$$(\mathbb{P}_n - \mathbb{P}) \left\{ \varphi(Z; \widehat{\mathbb{P}}) - \varphi(Z; \mathbb{P}) \right\}$$

arises due to double usage of the data and can be controlled via the following approaches.

Approaches to Control:

- **Donsker Conditions:** Limit the complexity of estimated functions by assuming they belong to Donsker classes (e.g., smooth parametric models or Sobolev classes).
- **Cross-Fitting:** Partition the sample into folds, estimating nuisance parameters on one fold and evaluating on another.

Consistency Requirement:

Definition

An estimated influence function $\hat{\varphi}$ is L_2-consistent if:

$$\|\hat{\varphi}(Z) - \varphi(Z)\|_2 \to_{\mathbb{P}} 0,$$

where $\|f\|^2 = \int f^2 dP$.

Donsker Classes and Cross-Fitting

Donsker Conditions:

Proposition

If $\hat{\varphi}$ is L_2-consistent and belongs to a Donsker class, then:

$$(\mathbb{P}_n - \mathbb{P})\left\{\varphi(Z;\widehat{\mathbb{P}}) - \varphi(Z;\mathbb{P})\right\} = o_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right).$$

Cross-Fitting:

Proposition

Given K folds $\{\mathcal{I}_1, \ldots, \mathcal{I}_K\}$ partitioning the data, the cross-fitted empirical process satisfies:

$$\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in\mathcal{I}_k}\left(\mathbb{P}_n^{(k)} - \mathbb{P}\right)\left(\varphi(Z,\widehat{\mathbb{P}}^{(-k)}) - \varphi(Z,\mathbb{P})\right) = o_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right)$$

Drawbacks: While cross-fitting reduces overfitting, it may introduce finite-sample bias due to reduced effective sample size in each fold.

Second-Order Remainder

The second-order remainder term:

$$R_2\left(\widehat{\mathbb{P}}, \mathbb{P}\right) = \left[\psi(\widehat{\mathbb{P}}) - \psi\left(\mathbb{P}\right)\right] - \mathbb{P}\left\{\varphi\left(Z, \widehat{\mathbb{P}}\right)\right\}$$

arises due to second-order products of differences between $\mathbb{P}_n$ and $\mathbb{P}$. It vanishes at a rate $o_{\mathbb{P}}(1/\sqrt{n})$ under certain conditions.

Example

Average Treatment Effect

$$\varphi(Z; \mathbb{P}) = \frac{\mathbf{1}(T = t)}{\mathbb{P}(T = t \mid X)} \left(Y - \mu_t(X)\right) + \mu_t(X) - \psi,$$

with the remainder given by:

$$R_2(\bar{P}, P) = \int \left\{\frac{1}{\bar{\pi}_t(x)} - \frac{1}{\pi_t(x)}\right\} \{\mu_t(x) - \bar{\mu}_t(x)\} \pi_t(x) dP(x),$$

where $\pi_t(x) = P(T = t \mid X = x)$ and $\bar{\pi}_t(x) = \bar{P}(T = t \mid X = x)$.

Bound: Using Cauchy-Schwarz: $|R_2(\mathbb{P}_n, \mathbb{P})| \leq \left(\frac{1}{\epsilon}\right) \|\widehat{\pi}_t - \pi_t\| \|\widehat{\mu}_t - \mu_t\|$, ensuring $o_{\mathbb{P}}(1/\sqrt{n})$ if $\|\widehat{\pi}_t - \pi_t\| = o_{\mathbb{P}}(n^{-1/4})$ and $\|\widehat{\mu}_t - \mu_t\| = o_{\mathbb{P}}(n^{-1/4})$.

Summary

Key Insight: When the empirical process and remainder terms are of the order $o_{\mathbb{P}}(1/\sqrt{n})$, the sample average term S^* dominates, leading to:

$$\sqrt{n}(\widehat{\psi}^{os} - \psi) = \sqrt{n}S^* + o_{\mathbb{P}}(1) \rightsquigarrow N(0, \text{var}\{\varphi(Z; \mathbb{P})\}),$$

by the central limit theorem and Slutsky's theorem.

Implications:

- $\widehat{\psi}$ is root-n consistent and asymptotically normal.
- Asymptotically valid 95% confidence intervals for ψ can be constructed as:

$$\widehat{\psi} \pm 1.96\sqrt{\widehat{\text{var}}\{\varphi(Z; \widehat{\mathbb{P}})\}/n}.$$

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Benchmarks and Minimax Risk

Goal. Quantify the intrinsic difficulty of estimating ψ over a model $\mathcal{P}$.

Global minimax benchmark. A standard lower bound is the (squared-error) minimax risk

$$\inf_{\widehat{\psi}} \sup_{P \in \mathcal{P}} \mathbb{E}_P\left[(\widehat{\psi} - \psi(P))^2\right] \;\geq\; R_n.$$

- R_n is the smallest possible worst-case MSE over $\mathcal{P}$.
- For many nonparametric problems (e.g. density / regression estimation), R_n decays slower than $1/n$.
- For *functionals* $\psi(P)$, it is often possible to achieve $R_n \asymp 1/n$.

We now zoom in locally around a fixed P and study more refined *local asymptotic minimax* bounds.

Cramér–Rao Bound in Parametric Models

Setup. Let $\{P_\theta : \theta \in \mathbb{R}^p\}$ be a smooth parametric model, and $\psi(\theta)$ a smooth scalar functional.

- **Score:** $s_\theta(Z) = \nabla_\theta \log p_\theta(Z)$.
- **Fisher information:** $I_\theta = \mathrm{Var}_\theta\{s_\theta(Z)\}$.
- **Gradient:** $\dot{\psi}(\theta) = \nabla_\theta \psi(\theta)$.

Cramér–Rao bound (scalar ψ).

For any regular unbiased estimator $\widehat{\psi}$,

$$\mathrm{Var}_\theta(\widehat{\psi}) \;\geq\; \dot{\psi}(\theta)^\top \, I_\theta^{-1} \, \dot{\psi}(\theta).$$

Geometric view.

- Tangent space is $\mathbb{R}^p$ with inner product $\langle u, v \rangle_\theta = u^\top I_\theta v$.
- $\dot{\psi}(\theta)$ is the *direction* in which ψ changes.
- The CR bound is the squared norm of this direction in the Fisher metric: $\|\dot{\psi}(\theta)\|^2_{I_\theta^{-1}}$.

From Cramér–Rao to Local Minimaxity

Classical Cramér–Rao bound (CRB). For unbiased estimators in smooth parametric models:

$$\mathrm{Var}_\theta(\widehat{\psi}) \;\geq\; \dot{\psi}(\theta)^\top I_\theta^{-1} \dot{\psi}(\theta).$$

Limitations of CRB.

- Only applies to *unbiased* estimators.
- Gives a bound on the *variance*, not the full MSE.
- Pointwise in θ (not uniform over neighborhoods).
- Does not directly extend to semiparametric or NP models.

Le Cam's insight. Le Cam's Local Asymptotic Normality (LAN) theory shows that, in a shrinking neighborhood of θ, the estimation problem is equivalent to a 1d Gaussian shift experiment. This yields a *local asymptotic minimax* MSE bound, refining and strengthening CRB.

Theorem (Van der Vaart, 2000, Thm 8.11).

Assume:

- P_θ is differentiable in quadratic mean at θ, with nonsingular Fisher information $I_\theta = \mathrm{Var}_\theta\{s_\theta(Z)\}$,
- $\psi(\theta)$ is differentiable with gradient $\dot{\psi}(\theta)$.

Then for any estimator sequence $\widehat{\psi}$,

$$\inf_{\delta>0} \liminf_{n\to\infty} \sup_{\|\theta'-\theta\|<\delta} n\,\mathbb{E}_{\theta'}\left[(\widehat{\psi} - \psi(\theta'))^2\right] \;\geq\; \dot{\psi}(\theta)^\top I_\theta^{-1} \dot{\psi}(\theta).$$

Interpretation.

- In any shrinking neighbourhood of θ, the asymptotic local MSE cannot beat the CR bound.
- Regular MLE / one-step estimators with influence function $\varphi_\theta(Z) = \dot{\psi}(\theta)^\top I_\theta^{-1} s_\theta(Z)$ *attain* this optimal bound.

From Parametric to Nonparametric via Submodels

Idea. To derive lower bounds in a large (semi- or nonparametric) model $\mathcal{P}$, we restrict attention to *1-dimensional parametric submodels* passing through P. Any lower bound on such a submodel is automatically valid for the full model.

Standard nonparametric perturbation. For any h with $\|h\|_\infty \le M$, define the multiplicative path

$$p_\varepsilon(z) = p(z)\{1 + \varepsilon h(z)\}, \qquad \varepsilon < 1/M.$$

This construction defines a smooth curve $\varepsilon \mapsto P_\varepsilon$ with $P_0 = P$ and

- the score is simply

$$s(z) = \frac{\partial}{\partial \varepsilon} \log p_\varepsilon(z)\Big|_{\varepsilon=0} = h(z),$$

- the Fisher information along the path reduces to $\mathbb{E}_P[h(Z)^2]$,
- the Cramér–Rao bound along this submodel takes the explicit form

$$\frac{\{\partial_\varepsilon \psi(P_\varepsilon)|_{\varepsilon=0}\}^2}{\mathbb{E}_P[h(Z)^2]}.$$

We will now optimize this bound over all directions h in the tangent space.

Best Submodel Bound and the EIF Variance

Assume ψ is pathwise differentiable with efficient influence function $\varphi(Z; P)$.

Pathwise derivative along h. For the submodel $p_\varepsilon(z) = p(z)\{1 + \varepsilon h(z)\}$,

$$\frac{\partial}{\partial \varepsilon}\psi(P_\varepsilon)\Big|_{\varepsilon=0} = \int \varphi(z; P)\, h(z)\, dP(z) = \mathbb{E}\{\varphi(Z; P)h(Z)\}.$$

Cramér–Rao along this submodel.

$$\frac{\psi'(P_\varepsilon)^2}{\mathrm{Var}_{P_\varepsilon}\{s_\varepsilon(Z)\}}\Big|_{\varepsilon=0} = \frac{\mathbb{E}\{\varphi(Z; P)h(Z)\}^2}{\mathbb{E}\{h(Z)^2\}}.$$

Optimize over h.

$$\sup_h \frac{\mathbb{E}\{\varphi(Z; P)h(Z)\}^2}{\mathbb{E}\{h(Z)^2\}} \;\le\; \mathbb{E}\{\varphi(Z; P)^2\} = \mathrm{Var}\{\varphi(Z; P)\},$$

by Cauchy–Schwarz, with equality for $h \propto \varphi$ (when $\varphi \in \mathcal{T}(P)$).

Geometric intuition.

- Scores h are directions in the tangent space $\mathcal{T}(P)$.
- The map $h \mapsto \psi'(P)[h]$ is linear; its operator norm is $\|\varphi\|_{L_2(P)}$.
- The *best* CR bound over all submodels equals $\mathrm{Var}\{\varphi(Z; P)\}$.

Nonparametric Efficiency Bound

Theorem (van der Vaart, 2002, Corollary 2.6)

Let $\psi : \mathcal{P} \to \mathbb{R}$ be pathwise differentiable at P with efficient influence function $\varphi \in \mathcal{T}(P)$. Assume:

- the model is nonparametric, or the tangent space $\mathcal{T}(P)$ is a convex cone.

Then for any estimator sequence $\widehat{\psi}_n$,

$$\inf_{\delta>0} \liminf_{n\to\infty} \sup_{\mathrm{TV}(P,Q)<\delta} n\,\mathbb{E}_Q\left[(\widehat{\psi}_n - \psi(Q))^2\right] \;\geq\; \mathrm{Var}\{\varphi(Z;P)\}.$$

Implications.

- $\mathrm{Var}\{\varphi(Z;P)\}/n$ is the *nonparametric efficiency bound*.
- No estimator can have smaller local asymptotic MSE.
- Any estimator with

$$\sqrt{n}(\widehat{\psi}_n - \psi(P)) \rightsquigarrow N\big(0, \mathrm{Var}\{\varphi(Z;P)\}\big)$$

  is locally minimax optimal.

One-Step Estimator and Local Minimax Optimality

Recall. For our target ψ we constructed a one-step estimator $\widehat{\psi}^{os}$ based on the efficient influence function φ:

$$\widehat{\psi}^{os} = \psi(\widehat{P}) + P_n\{\varphi(Z; \widehat{P})\},$$

with cross-fitting to control empirical process terms.

Asymptotic distribution. Under our regularity conditions,

$$\sqrt{n}\big(\widehat{\psi}^{os} - \psi(P)\big) \rightsquigarrow N\big(0,\, \mathrm{Var}\{\varphi(Z; P)\}\big).$$

Conclusion.

- $\widehat{\psi}^{os}$ attains the nonparametric efficiency bound $\mathrm{Var}\{\varphi(Z; P)\}/n$.
- In the local asymptotic minimax sense, no other estimator can be uniformly better.
- Geometrically: we are using the *canonical gradient* direction φ to correct the plug-in estimator in the most efficient way.

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Why Many Functionals Have a Parametric $1/n$ Rate

Function vs. functional. Estimating an infinite-dimensional object (e.g. a density or regression function) in dimension d and smoothness s suffers from the curse of dimensionality:

$$R_n(\text{function}) \asymp n^{-2s/(2s+d)}.$$

(d = dimension of the input space, s = Hölder/Sobolev smoothness of the target function.)

But estimating a smooth scalar functional is often much easier. Examples:

$$\psi(P) = \mathbb{E}[g(Z)], \qquad \psi(P) = \mathbb{E}[m(X)], \qquad \psi(P) = \text{ATE}.$$

These depend on P only through a *single direction* in the tangent space, so typically:

$$R_n(\psi) \asymp \frac{1}{n}.$$

Le Cam's insight. In a local asymptotic sense, a smooth functional behaves like a *one-dimensional parameter* inside an infinite-dimensional model. Its optimal variance is

$$\frac{1}{n} \operatorname{Var}\{\varphi(Z)\}.$$